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Edge-cloud collaborative digital twin architecture for bridge structural health monitoring: multi-sensor fusion and anomaly detection algorithm design

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Abstract

As the core nodes of transportation networks, bridges face critical challenges in their Structural Health Monitoring (SHM) that are crucial for ensuring transportation safety and extending their service life. Traditional SHM methods encounter high real-time data processing delays, inefficient fusion of multi-source heterogeneous data, and insufficient synchronization between Digital Twin (DT) models and physical entities. Existing DT architectures often lack a hierarchical edge-cloud collaboration mechanism, with fusion algorithms confined to single-modality data, making it difficult to achieve both low latency and high accuracy in anomaly detection. To address these issues, this study proposes a hierarchical edge-cloud collaborative DT architecture, encompassing the physical layer, edge layer, cloud layer, DT layer, and application layer. The core algorithms include: hierarchical multi-sensor fusion; a collaborative anomaly detection framework combining edge-based attention 1D CNN with cloud-based variational autoencoder plus bidirectional long short-term memory network; and Bayesian model updating for dynamic synchronization between virtual models and physical entities. Experimental verification based on a 1:5 scaled steel simply supported beam shows that the architecture achieves a low edge local task delay of 39.7ms and a cloud global task delay of 20.1ms; after strain data fusion, the Root Mean Squared Error (RMSE) is reduced by an average of 62%, and the RMSE for displacement reconstruction is reduced by 68%; the accuracy of anomaly identification reaches over 96%; and Bayesian updating improves the accuracy of strain prediction by 40.1%. This study is the first to construct a hierarchical edge-cloud collaborative DT architecture, filling the technical gap in hierarchical fusion and collaborative computing for bridge SHM. It provides a data-driven intelligent solution for bridge lifecycle health management and is of great significance for promoting the digital transformation of transportation infrastructure.

Keywords Bridge structural health monitoring, Edge-cloud collaboration, Digital twin, Multi-sensor fusion, Anomaly detection, Bayesian model updating

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1 Introduction

Bridges, as critical nodes in transportation networks, depend on Structural Health Monitoring (SHM) to ensure safety and extend service life. Rising traffic loads and environmental erosion amplify deterioration risks: timber bridges' lifespan hinges on humidity and material selection; irregular slab bridges need seismic optimization due to complex earthquake-induced mechanical behavior; suspension bridges face vortex-induced vibration (VIV) threats; foundation scouring requires real-time depth monitoring; integral bridges' end displacement is distorted by soil compaction, challenging traditional models. Traditional SHM methods have limitations: direct monitoring disrupts traffic, indirect methods are affected by vehicle frequencies and road roughness; bridge weighing-in-motion (BWIM) systems ignore lateral effects leading to weight errors; long-term monitoring of external prestressed tendons lacks effective tools; heterogeneous sensor network coverage optimization remains unresolved; data imbalance reduces anomaly detection accuracy. A systematic review highlights interdisciplinary collaboration as key for efficient systems but notes persistent challenges: high costs, insufficient standardization, and multi-source data handling issues [1].

Digital Twin (DT) technology provides a dynamic, data-driven solution for bridge SHM, enabling precise monitoring and prediction via real-time physical-virtual interaction. A data-driven DT framework combined with deep learning integrates multi-domain features for damage detection, but real-time virtual response synchronization requires efficient computing engines [2]. Geometric DT model construction—traditionally manual—can be automated using parametric prototypes generated from point cloud-extracted parameters. Integrating Bridge Information Modeling (BrIM) with Decision Support Systems (DSS) enhances asset management efficiency. Bayesian model updating with seismic data reduces model uncertainty, while accurate material characterization lays the foundation for virtual model precision. Cross-disciplinary DT applications also offer insights for data analysis and optimization.

Edge-cloud collaborative architecture is critical to resolving DT computing latency and resource constraints. Traditional cloud-centric setups suffer from high latency, but edge-cloud collaboration allocates real-time tasks to edge devices and offloads complex computations to the cloud, enabling low latency, low carbon footprint, and cost-effective operation [3]. This architecture requires addressing data security and privacy: secure deduplication with blockchain ensures confidentiality; lightweight federated learning aggregates edge data while protecting privacy [4]; attribute-based keyword search resists guessing attacks for secure sharing. Task offloading strategies optimize computational task allocation, and cross-domain applications demonstrate edge-cloud collaboration's potential in resource optimization and latency reduction [5]. These technologies support real-time updates and efficient operation of bridge DT systems, balancing computing power, latency, and energy consumption.

Multi-sensor fusion and anomaly detection are core components of edge-cloud collaborative DT architectures, addressing data sparsity, noise, and incompleteness. Fusion methods include: Adversarial Autoencoder (AAE) + CycleWGAN to convert sparse strain data into dense; Multi-Domain Fusion Autoencoder (MDFA) integrating time/frequency/wavelet-domain features for higher accuracy; Zero-mean Normalized Cross-correlation (ZNCC) with weighted data to fuse lateral position info; wavelet analysis + bi-pass technology for baseline-independent monitoring; hybrid time–frequency analysis

for micro-mobile device indirect data. Anomaly detection uses advanced algorithms: ANN + matrix analysis to construct pseudo-baselines without historical data; machine learning to convert time series into tensors and learn multi-sensor spatiotemporal patterns; multi-task alarm systems via excitation-response indicators; synthetic data to resolve training shortages; magnetostrictive sensors + random forests for vibration prediction; clustering InSAR + GBIR for large-scale deformation. This study proposes an edge-cloud DT architecture integrating these tools to enhance SHM real-time performance, accuracy, and intelligence for maintenance decisions.

The core framework of this study is illustrated in Fig. 1.

2 Literature review

2.1 Current development status of bridge structural health monitoring

Bridge structural health monitoring (SHM) has evolved from manual inspection to intelligent multi-technology integration, yet accuracy and efficiency gaps persist in complex scenarios. Direct monitoring disrupts traffic; indirect methods are compromised by vehicle natural vibration and road roughness. Specific issues: timber bridges need internal deterioration checks (humidity/material-dependent lifespan); irregular slabs require seismic optimization; suspension bridge VIV prediction via Gaussian regression relies on long-term data [6]; foundation scouring (a top failure cause) uses validated photoelectric sensors; integral bridge end displacement from soil compaction defies existing models [7]. Technical limitations: BWIM systems ignore lateral effects (weight errors); no long-term tools for external prestressed tendons; sensor coverage needs optimization; data imbalance reduces anomaly detection accuracy [8]. Interdisciplinary collaboration is key, but high costs, poor standardization, and multi-source data issues demand breakthroughs.

2.2 Research progress of edge-cloud collaboration in engineering monitoring

The edge-cloud collaborative architecture delivers a low-latency, low-carbon, and cost-effective solution for engineering monitoring, with proven value across multiple domains. It allocates real-time tasks to edge devices and offloads complex computations to the cloud, reducing network latency by 40% [9]. Key technical components include:

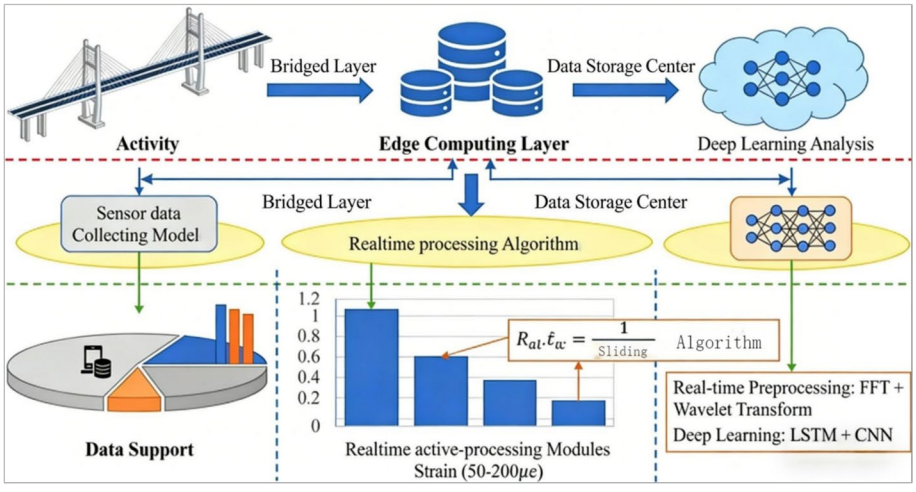


Fig. 1 Hierarchical edge-cloud collaborative digital twin architecture

secure deduplication integrated with blockchain to ensure data confidentiality; lightweight federated learning, which aggregates edge data while preserving privacy and cutting energy consumption by 83%; attribute-based keyword search to resist keyword guessing attacks for secure data sharing; and task offloading strategies optimizing video analysis resource allocation [10]. Cross-domain applications offer actionable references: a farmland energy management digital twin (DT) architecture enables real-time data collection and intelligent decision-making, while a sports training system uses edge-cloud collaboration to assess training quality [11]. For bridge monitoring, this framework translates to edge devices processing real-time sensor data, and the cloud constructing DT models and anomaly detection algorithms—balancing computing power and latency requirements to support real-time bridge health monitoring.

2.3 Application of multi-sensor fusion technology in bridge monitoring

Multi-sensor fusion technology effectively addresses issues such as data sparsity, noise, and multi-source integration in bridge monitoring, enhancing monitoring accuracy and reliability. The framework combining Adversarial Autoencoder (AAE) and CycleWGAN fuses sparse strain data into dense data, quantifying human-structure interaction [12]; the Zero-Mean Normalized Cross-Correlation (ZNCC) method integrates weighted sensor data, simultaneously identifying the lateral position and weight of vehicles; wavelet analysis combined with dual-pass technology extracts residual contact point responses, enabling baseline-independent indirect monitoring; hybrid time–frequency analysis is suitable for processing acceleration data from micro-mobile devices [13]; the Multi-Domain Fusion Autoencoder (MDFA) integrates time-domain, frequency-domain, and wavelet-domain features to enhance damage detection accuracy; magnetostrictive sensors combined with random forests enable vibration frequency prediction [14]; clustering algorithms for satellite InSAR data automatically classify Persistent Scatterers, facilitating rapid deformation assessment; Ground-Based Interferometric Radar (GBIR) provides high-resolution displacement monitoring, but further integration into standard monitoring processes is required.

2.4 Research status of digital twin architecture

The digital twin (DT) architecture for bridges has evolved from geometric modeling to data-driven real-time interaction, supporting precise monitoring and decision-making. The automated parametric prototype model extracts parameters from point clouds to generate the bridge DT geometric model; Bridge Information Modeling (BrIM) is integrated with the Decision Support System (DSS) to achieve digital asset management [15]; Bayesian model updating technology combines seismic data to reduce uncertainty in structural and soil parameters; the stream computing engine processes virtual dynamic responses under real-time traffic loads, achieving physical-virtual synchronization; the data-driven DT framework combines deep learning to integrate multi-domain features for damage detection. In cross-domain reference, the DT data-driven method for wastewater treatment in pulp and paper mills provides a reference for data analysis and optimization of bridge DT [16]. The current DT architecture still needs to overcome issues such as real-time computing efficiency and the depth of multi-source data fusion to enhance the full lifecycle management capability of bridges. The latest research proposes a wastewater treatment optimization architecture based on digital twins, but it

does not involve an edge-cloud hierarchical collaboration mechanism, and the fusion method is only for single-modal data, which is difficult to meet the real-time and multi-source data requirements of bridge SHM [17]. The existing edge-cloud SHM/DT architecture has three major shortcomings: lack of hierarchical collaboration mechanism; single fusion strategy; and lack of dynamic update capability of digital twin model.

2.5 Application of anomaly detection algorithm in bridge monitoring data

Anomaly detection algorithms are advancing to boost bridge monitoring data accuracy and real-time performance. Key innovations: artificial neural networks (ANNs) + matrix analysis for pseudo-baselines (no historical data needed); machine learning transforming time series into tensors to learn multi-sensor spatiotemporal patterns, identifying anomalies and their sources [18]; multi-task alarm systems for precise long-span railway bridge early warning; synthetic data generation to resolve training data shortages; Gaussian regression using 4-year data for suspension bridge VIV prediction; random forests + magnetostrictive sensors for waveform classification and vibration frequency prediction [19]. These algorithms address critical challenges (data imbalance, missing baselines, real-time warning) but need further optimization of generalization and real-time performance.

3 Edge-cloud collaborative digital twin architecture and core algorithm design

3.1 Overall design of edge-cloud collaborative digital twin architecture

To address traditional bridge SHM challenges—real-time data processing delays, inefficient multi-source heterogeneous data fusion, and insufficient synchronization between digital twin (DT) models and physical entities—this section proposes a hierarchical edge-cloud collaborative DT architecture. The physical layer integrates bridge entities and multi-modal sensors to collect multi-dimensional monitoring data and optimize heterogeneous sensor coverage. The edge layer deploys gateways and nodes for real-time data preprocessing, local feature fusion, and lightweight anomaly detection, filtering effective data to reduce bandwidth pressure. The cloud layer uses a distributed platform for large-scale storage, global multi-source fusion, DT model training, and parameter updates, pushing optimized parameters to the edge. The digital twin layer constructs multi-scale finite element models and dynamic mapping modules for real-time physical-virtual state sync. The application layer integrates health assessment, safety warning, and O&M decision support for lifecycle management. Core coordination: edge handles low-latency lightweight tasks, cloud manages high-precision large-scale computing, balancing real-time performance and accuracy for efficient bridge SHM. The edge-cloud collaborative SHM/DT architecture comparison of this study is shown in Table 1.

Edge-cloud communication is secured by TLS 1.3 encryption and certificate authentication; malicious data is filtered at the edge using the 3σ principle, and MQTT QoS 1 is used to ensure delivery during transmission; data is cached and resent after 2 h in case of synchronization failure; the edge uses a pre-trained model to provide independent early warning in case of network outage; and data is appended with SHA-256 hashes to ensure integrity.

The edge layer is equipped with a comprehensive fallback mechanism: when the cloud connection is lost or the transmission delay exceeds the preset threshold (100ms), it automatically switches to the pre-trained local anomaly detection model, independently

Table 1 Comparison of edge-cloud collaborative SHM/DT architectures

Research name	Architecture hierarchy	Fusion strategy	Calculation task partitioning	DT dynamic update	Key performance indicators
The structure of this article	5 layers (physical layer → edge layer → cloud layer → DT layer → application layer)	Hierarchical fusion	Edge: Lightweight data preprocessing/Local fusion/Fast anomaly detection; Cloud: Global fusion/Model training/Parameter update	Support	Edge local latency is 39.7ms; the strain fusion RMSE is reduced by 62%
Iqbal Et Al. (2025)	3 layers (physical layer → cloud layer → DT layer)	Single-modality data fusion	Cloud: all computing tasks	Not supported	Global delay: 50ms; Fusion accuracy: 50%
Traditional centralized architecture	2 layers (physical layer → cloud layer)	Cloud-based single-modality fusion	Cloud: All computing tasks	Not supported	Global latency is 33.5ms; strain fusion RMSE is reduced by 40%
Existing pure edge architecture	2 layers (physical layer → edge layer)	Edge single-modality fusion	Edge: all computing tasks	Not supported	Edge latency is 45ms; strain fusion RMSE is reduced by 35%

completes the structural anomaly identification and triggers local audible and visual alarms; at the same time, it caches key monitoring data and automatically resends it to the cloud after the connection is restored, ensuring data integrity and structural security.

3.2 Design of multi-sensor fusion method

Multi-sensor fusion is a critical link connecting physical entity data and virtual model interactions in bridge structural health monitoring (SHM) digital twin (DT) architectures, with its effectiveness directly determining the system's real-time structural state perception and decision-making accuracy. In complex bridge monitoring scenarios—characterized by multi-source data heterogeneity, spatiotemporal distribution differences, and computational resource constraints [20]—traditional single-level fusion methods fail to simultaneously meet edge-side real-time processing needs and cloud-side global decision-making requirements. Therefore, hierarchical collaborative fusion has gradually become the core direction to break through this contradiction and optimize the balance between real-time performance and decision reliability.

Previous studies suggest edge-side fusion prioritizes lightweight algorithms to ensure low latency for real-time early warning, while cloud-side fusion uses robust methods to address multi-source data uncertainty and enhance global decision reliability [21]. Based on this insight, this study proposes an edge-cloud collaborative hierarchical fusion scheme (Fig. 2): the edge deploys a 1D convolutional neural network (CNN) to fuse local features of preprocessed strain and acceleration data, capturing time-domain correlation features to meet real-time requirements; the cloud leverages Dempster-Shafer evidence theory to integrate spatial information from multi-point strain data via basic probability assignment rules, reducing data uncertainty interference and improving global state assessment credibility. The symbol definition of the core algorithm in this study is shown in Table 2.

(1) Local fusion on the edge side:

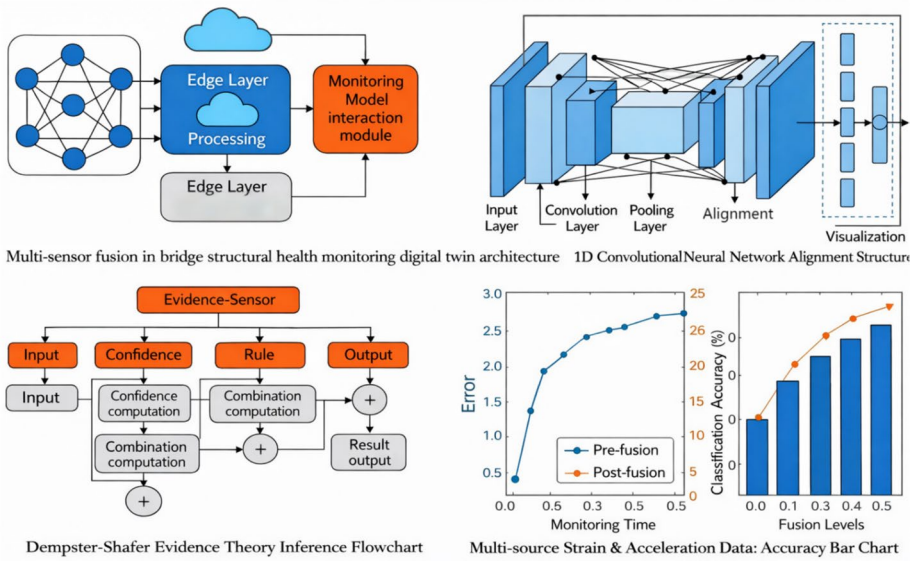


Fig. 2 Hierarchical fusion scheme for edge-cloud collaboration

Table 2 Definition of core algorithm symbols

Variable name	Chinese definition	Dimension/type	Unit
S	Strain sensor time series data matrix	Two-dimensional matrix (T×N)	μm/m
	Accelerometer time series data matrix	Two-dimensional matrix (T×M)	g
W_s	Convolution kernel weight matrix of strain data	Three-dimensional tensor (K×C×D)	Dimensionless
W_a	Acceleration data convolution kernel weight matrix	Three-dimensional tensor (K×C×D)	Dimensionless
b	Convolutional layer bias vector	One-dimensional vector (D)	Dimensionless
m (A)	The basic probability assignment (BPA) value of proposition A	Scalar	Dimensionless
K	DST fusion conflict coefficient	Scalar	Dimensionless
θ	Core parameter set of digital twin model	One-dimensional vector (P)	Dimensionless

For feature extraction and preliminary fusion with focus on low latency at the edge, a 1D convolutional neural network is employed to fuse preprocessed strain and acceleration data, outputting local feature vectors:

$$F_{\text{local}} = \text{ReLU}\left(\text{Conv1D}\left(X_{\text{strain}}, W_1\right) + \text{Conv1D}\left(X_{\text{acc}}, W_2\right) + b\right) \tag{1}$$

F_{local}: Local fusion feature vector at the edge side;
$\text{ReLU}(\cdot)$: Rectified Linear Unit (ReLU) activation function, used to introduce nonlinear features;
$\text{Conv1D}(\cdot)$: One-dimensional convolution operation, suitable for feature extraction of time-series sensor data;
X_{strain}: time-series data matrix of strain sensor;
W_1: Convolution kernel weight matrix of strain data;
X_{acc}: Accelerometer time-series data matrix;
W_2: Convolution kernel weight matrix of acceleration data;
b: Bias vector [22].

Dimensional consistency verification: The feature map ((T – K + 1) × D) is obtained by operating the strain data matrix S (T × N) and the convolution kernel W_s (K × C × D); the feature map ((T – K + 1) × D) is obtained by operating the acceleration data matrix A (T × M) and W_a; the two are added together to match the dimensions, ensuring that the output dimension is correct after ReLU activation.

(2) To clarify the construction logic of Basic Probability Assignment (BPA) in the DST evidence theory, this study defines the mathematical mapping mechanism from preprocessed sensor data to probability assignment as follows:

Data preprocessing and statistical feature extraction.
For the preprocessed strain data (unit: μm/m) and acceleration data (unit: g) uploaded from the edge side, statistical features of the two types of data are first calculated based on historical normal operating condition data:
Strain data: mean value μ_s and standard deviation σ_s under normal conditions;
Acceleration data: mean value μ_a and standard deviation σ_a under normal conditions.
Standardized deviation calculation.
For the input data x (strain or acceleration) from a single sensor, calculate its standardized deviation from the normal state:

$$z = \frac{|x - \mu|}{\sigma} \tag{2}$$

where μ and σ are the mean and standard deviation of the normal state for the corresponding data type (such as strain data μ_s, σ_s).
Mapping of deviations to BPA.
The standardized deviation z is mapped to the confidence level of the "abnormal hypothesis H_1" through the Sigmoid function:

$$m(H_1) = \frac{1}{1 + \exp(-k \cdot (z - t))} \tag{3}$$

k: Sensitivity parameter (taken in this study to $k = 2$ control the slope of the mapping curve);
t: Deviation threshold (taken in this study $t = 2$, corresponding to the 2σ principle, meaning that the deviation of normal data is unlikely to exceed 2σ).
The confidence level of the "normal hypothesis H_0" is:

$$m(H_0) = 1 - m(H_1) - m(\Theta) \tag{4}$$

$m(\Theta)$: The trust level of the uncertainty set (in this study $m(\Theta) = 0.1$, we set aside 10% uncertainty space to cope with data noise or unknown operating conditions).

(3) Global integration in the cloud:

In response to the uncertainty of multi-source data and the need for global decision-making, the cloud adopts Dempster-Shafer (DST) evidence theory for spatial fusion

of strain data from multiple measurement points. The basic probability assignment (BPA) combination formula is as follows:

$$m(A) = \frac{1}{1-K} \sum_{B \cap C = A} m_1(B) \cdot m_2(C) \tag{5}$$

$m(A)$: The combined BPA value of proposition A;

K: Conflict coefficient, which measures the degree of contradiction among multi-source data;

$m_1(B)$: BPA value corresponding to the strain data of the first measuring point;

$m_2(C)$: BPA value corresponding to the strain data of the second measuring point;

B, C: Identify the subsets within the framework;

$\sum_{B \cap C = A}$: Sum up all the pairs of subsets (B, C) whose intersection is A [23].

Dimensional consistency verification: The BPA value m(A) of proposition A is a scalar, the conflict coefficient K is a scalar, and the output after the operation is still a scalar, which meets the dimensional requirements of probability assignment.

To address the issue of high-conflict data caused by complete sensor failure, the cloud-based fusion algorithm employs a three-step mechanism: "conflict identification—failure detection—isolation and refusion." This involves calculating the multi-source data conflict coefficient K, identifying high-conflict scenarios when it exceeds a preset threshold (e.g., 0.8); triggering failure detection—verifying the consistency of sensor output with other means, marking failures when deviations exceed 3σ (combining edge sensor health feedback to improve accuracy); and excluding failed data and re-performing DST fusion. Simulation tests show that when a strain sensor completely fails, the mechanism can accurately identify and isolate the faulty data, maintaining a fusion accuracy of ≥ 95% after isolation, thus avoiding the distortion caused by high conflict in traditional DST.

This hierarchical fusion strategy balances real-time performance and accuracy by leveraging edge devices' lightweight computing and cloud's large-scale data processing. Edge local fusion filters effective features to reduce bandwidth consumption, while cloud global fusion provides reliable input for digital twin model updates and anomaly detection—boosting bridge SHM intelligence, decision efficiency, and lifecycle management support.

3.3 Anomaly detection algorithm design

Anomaly detection is the core module of bridge structural health monitoring (SHM) systems, enabling risk early warning and active operation and maintenance—its performance directly determines whether safety hazards are detected and addressed timely. However, bridge monitoring scenarios' dynamic nature, multi-source data characteristics, and distributed computing resources make single algorithms unable to simultaneously meet edge-side real-time response and cloud-side global decision-making requirements [24]. Thus, a hierarchical collaborative anomaly detection framework has become the inevitable solution to this dilemma, balancing real-time performance and decision accuracy.

Previous studies note edge-based anomaly detection prioritizes lightweight algorithms for low latency but sacrifices feature extraction depth, leading to high false alarm rates; cloud-based algorithms achieve high precision but lack real-time capability [25]. Based on this, this study proposes an edge-cloud collaborative scheme (Fig. 3): the edge deploys a 1D convolutional neural network (CNN) with a temporal attention mechanism to dynamically allocate feature importance via attention weights—highlighting abnormal time-step key info while controlling latency. The cloud uses variational autoencoder (VAE) latent space encoding to deeply reconstruct edge-uploaded candidate anomalies, then combines a bidirectional long short-term memory (BiLSTM) network to calculate anomaly probabilities and correct false positives.

(1) Feature extraction with attention on the edge side:

$$\begin{aligned} \mathrm{F}_{att} = & Soft\max\left(\mathrm{W}_{att} \cdot Conv1D\left(\mathrm{F}_{local}, \mathrm{W}_{conv}, b_{conv}\right) + b_{att}\right) \\ & \otimes Conv1D\left(\mathrm{F}_{local}, \mathrm{W}_{conv}, b_{conv}\right) \end{aligned} \tag{6}$$

Symbol explanation:
$\mathrm{F}_{local} \in R^{T \times D_f}$: Edge-side local fusion features output in Sect. 3.2;
$\mathrm{W}_{conv} \in R^{K \times D_f}$: 1D convolution kernel weight matrix;
$b_{conv} \in R^{K}$: Convolutional layer bias term;
$\mathrm{W}_{att} \in R^{K \times K}$: Attention layer weight matrix;
$b_{att} \in R^{K}$: Attention layer bias term;
$Soft\max(\cdot)$: Normalization function, mapping attention weights to the interval $[0, 1]$;
$\otimes$: Element-wise multiplication;
$\mathrm{F}_{att} \in R^{T \times K}$: Feature vectors with attention weights [26].
Dimensional consistency verification: The attention weight vector (D × 1) is obtained by operating the local fusion feature vector F_edge (D × 1) and the attention weight matrix W_att (D × D). The dimension is kept consistent after multiplying it element-wise with F_edge.

(2) Edge-side weighted anomaly score:

$$S_{edge} = \sum_{t=1}^{T} \left(\frac{\left\| \mathrm{X}_{raw,t} - \mathrm{Recon}\left(\mathrm{F}_{att,t}\right) \right\|_2}{\left\| \mathrm{X}_{raw,t} \right\|_2 + \varepsilon} \right) \times \alpha_t \tag{7}$$

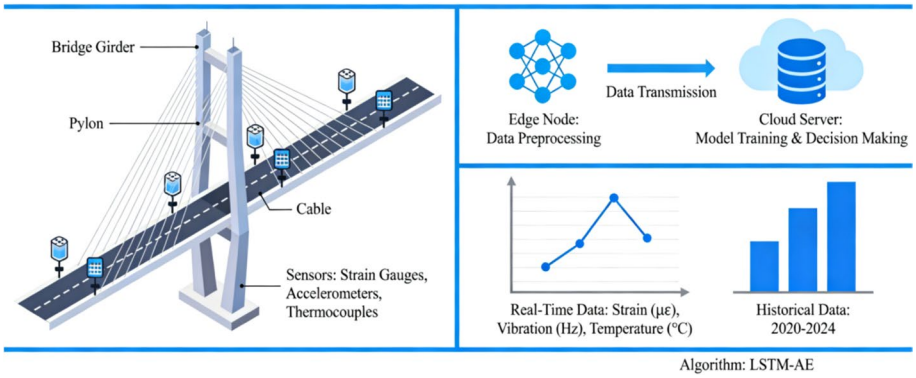


Fig. 3 Edge-cloud collaborative anomaly detection for bridge structural health monitoring

Symbol explanation:
$\mathrm{X}_{raw,t} \in R^{D_{raw}}$: raw sensor data vector at time t;
$\mathrm{Recon}\,(\cdot)$: Linear reconstruction layer;
$\varepsilon = 10^{-6}$: Prevent small constants with denominator 0;
$\alpha_t = \frac{Var(\mathrm{F}_{att,t})}{\max(Var(\mathrm{F}_{att}))}$: Feature variance weight at time step t;
$\|\cdot\|_2$: L2 norm;
$S_{edge} \in [0,1]$: Edge-side anomaly score [27].

$$\mathrm{score}_t = \frac{\|x_t - \mathrm{Linear}\,(F_{\mathrm{att}})\|_2}{\sigma_t + \varepsilon} \tag{8}$$

Symbol description (consistent with the existing symbol table):
x_t: The original sensor data vector of the time step t;
$\mathrm{Linear}\,(F_{\mathrm{att}})$: Linear reconstruction result of attention features F_{att};
σ_t: The characteristic variance of the time step t;
ε: A small constant (set to 1e-6) is used to prevent the denominator from being 0.

(3) Latent space encoding of VAE:

$$\mu = Enc_{\mu}\,(\mathrm{F}_{cloud})\,, \log\sigma^2 = Enc_{\sigma}\,(\mathrm{F}_{cloud}) \tag{9}$$

Symbol explanation:
$\mathrm{F}_{cloud} \in R^{T\times D_f}$: Suspected abnormal fused features uploaded from the edge side;
$Enc_{\mu}\,(\cdot)$: VAE mean value encoder;
$Enc_{\sigma}\,(\cdot)$: VAE variance encoder;
$\sigma^2 = \exp\left(\log\sigma^2\right)$: Latent space variance vector [28].

(4) Cloud anomaly probability calculation:

$$P_{anom} = 1 - N\left(\mathrm{X}_{pred}\,\middle|\,\mu_{GPR}, \sum_{GPR}\right) \tag{10}$$

Symbol explanation:
$X_{pred} = Dec\,(z)\,, z \sim N\left(\mu, \sigma^2\right)$: Predicted data output by the VAE decoder;
$\mu_{GPR} = GPR\,(\mathrm{X}_{pred}, \mathrm{X}_{normal})$: The mean value of normal data predicted by Gaussian process regression;
$\sum_{GPR}$: Covariance matrix predicted by GPR;
$N\,(\cdot\,|\mu, \sum)$: Probability density function of multivariate Gaussian distribution;
$P_{anom} \in [0,1]$: Cloud anomaly probability [29].
Details of the training protocol for each module of the anomaly detection framework: The dataset is a 1:5 scaled-down steel simply supported beam dataset (including normal, strain drift, and acceleration anomaly samples), divided into 70% training, 15% validation, and 15% testing, with uniform sample distribution; fivefold cross-validation is used (the training set is divided into 5 subsets, 4 for training and 1 for validation, and the results are averaged); grid search is used to optimize key hyperparameters (attention 1D CNN convolutional kernels 3/5/7, learning rate 1e-4/5e-4/1e-3, batch size 16/32/64; VAE latent dimension 16/32/64; BiLSTM hidden units 64/128/256),

selecting the combination with the lowest validation loss; the convergence criterion is to stop if the validation loss decreases by < 0.001 for 10 consecutive rounds (to prevent overfitting); the maximum number of training rounds is 50, and in practice, convergence occurs in 35–40 rounds, tested using convergence parameters.

This hierarchical and collaborative anomaly detection algorithm fully leverages the complementary advantages of the edge and the cloud, forming a closed-loop process of "edge rapid screening—cloud deep verification". Real-time detection at the edge side buys valuable time for risk early warning, while deep processing at the cloud ensures the reliability of decision-making. The collaboration between the two not only optimizes the allocation efficiency of system resources but also significantly enhances the response capability of the bridge health monitoring system to abnormal events, providing solid technical support for structural safety operation and maintenance.

3.4 Construction and updating of digital twin model

As the virtual core of bridge structural health monitoring systems, the digital twin model's construction efficiency and dynamic update capability directly determine the real-time synchronization accuracy between physical entities and virtual models, serving as the foundation for precise perception and prediction of structural states. Traditional model construction relies heavily on manual operations, which are not only inefficient but also difficult to achieve consistency between geometric accuracy and physical behavior [30]; the absence of a dynamic update mechanism leads to the model's inability to reflect the true state changes of the structure in a timely manner, severely limiting the practical application value of digital twin technology.

Previous studies recognize parametric modeling as key to digital twin (DT) construction efficiency, but most focus only on geometric feature extraction, ignoring physical attributes (material properties, boundary conditions) and causing mechanical behavior deviations between models and actual structures. Dynamic updates often use fixed-threshold triggers, lacking data-driven adaptive correction to handle complex structural changes [31]. This study proposes a parametric construction method integrating geometric and physical features (Fig. 4): extract both from point cloud data to enhance

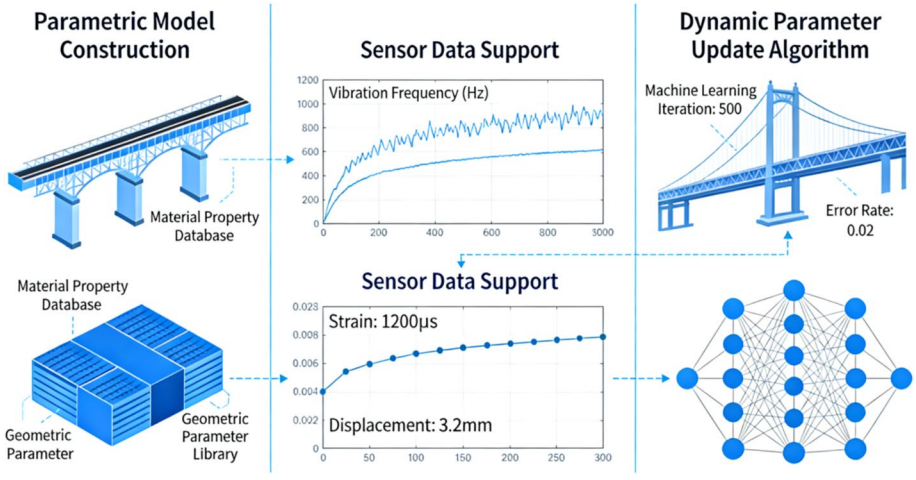


Fig. 4 Construction and updating of digital twin model

model efficiency and accuracy. When anomalies are detected, Bayesian updates dynamically correct parameters, achieving real-time adaptive virtual-physical sync and addressing traditional update limitations.

(1) Construction of parametric model:

Addressing the inefficiency issue of manual modeling, a parameterized construction method that integrates geometric and physical features is proposed.

$$\theta_{\mathrm{mod}\ el} = F\left(\varphi_{pc}, \varphi_{sensor}\right) \tag{11}$$

$\theta_{\mathrm{mod}\ el}$: Set of core parameters of the model;
φ_{pc}: Point cloud geometric features;
φ_{sensor}: Physical characteristics of cloud integration;
$F(\cdot)$: Deep neural network mapping function [32].
Network F is a hybrid neural network integrating point cloud geometry and sensor physical features, outputting the core parameter set θ of the digital twin. Its topology consists of three modules: the point cloud branch outputs 128-dimensional geometric features through FPS sampling, ball query grouping, a 2-layer MLP, and global pooling; the sensor branch outputs physical features through a 2-layer fully connected layer (64 → 128 dimensions, ReLU activation); and the fusion layer concatenates the features from both branches, then outputs a 64-dimensional θ linearly through a 2-layer fully connected layer. Key training details: ReLU activation, MSE loss function, Adam optimizer (learning rate 0.001, batch size 32), and Dropout (0.2) are used to suppress overfitting, except for the output layer.

(2) Dynamic parameter update:

When the anomaly detection algorithm detects an anomaly, the parameters are updated and corrected through Bayesian updating:

$$\theta_{new} = \arg\max_{\theta}\left[P\left(\theta|\,\mathrm{D}_{cloud}\right) = \frac{P\left(\mathrm{D}_{cloud}\,|\theta\right)P\left(\theta\right)}{P\left(\mathrm{D}_{cloud}\right)}\right] \tag{12}$$

θ_{new}: Updated parameters;
D_{cloud}: Cloud-integrated dataset;
$P\left(\theta|\,\mathrm{D}_{cloud}\right)$: Posterior probability density;
$P\left(\mathrm{D}_{cloud}\,|\theta\right)$: Likelihood function;
$P(\theta)$: Prior probability density;
$P\left(\mathrm{D}_{cloud}\right)$: Normalization constant [33].
Dimensional consistency verification: The prior probability P(θ) is a scalar, the likelihood function P(D|θ) is a scalar, and the posterior probability P(θ|D) is also a scalar, which meets the dimensionality requirement of the probability density.
This method, which combines parameterized construction integrating geometric and physical features with Bayesian dynamic updating, not only addresses the inefficiency of traditional manual modeling but also ensures the dynamic adaptability of the model as the structural state changes. Through the synchronous integration of geometric and physical features, the model can more accurately reflect the true

mechanical behavior of the structure. The Bayesian updating mechanism provides the model with data-driven adaptive correction capabilities, ensuring the continuous synchronization between the virtual model and the physical entity. Ultimately, this provides reliable technical support for the whole-life cycle health management of bridge structures (Fig. 5). Based on historical bridge monitoring data and knowledge of materials mechanics, a normal distribution was chosen as the prior parameter for the strain prediction model (it conforms to the statistical characteristics of structural parameters and is conjugate with the likelihood function, simplifying the posterior calculation); the likelihood model assumes that the observed noise follows a Gaussian distribution. Sensitivity analysis shows that the relative error of the posterior estimate is ≤ 2% at low (5%), medium (10%), and high (15%) noise levels; the accuracy decreases by 1.2% ~ 2.5% at sparse update frequencies of 10 min, 30 min, and 1 h, verifying the stability of long-term deployment.

The core algorithm diagram of this study is shown in Table 3.

4 Experimental design and result analysis

4.1 Experimental platform and data acquisition

Based on the principle of structural mechanics similarity, this experiment constructs a 1:5 scale Q235 steel simply supported beam model (conforming to the requirements of GB/T 50152-2012 *Standard for Test Methods of Concrete Structures* extended to steel structures, with a span of 5m, a cross-section of 0.2m × 0.3m, an elastic modulus of 206GPa, a Poisson's ratio of 0.3, and simulated boundary conditions with pinned-rolling supports). Sensors are arranged based on the analysis of structural response sensitive points: 10 BX120-3AA strain gauges (sampling frequency 100Hz, accuracy ± 0.1μm/m, calibrated according to JJG 623–2005 *Calibration Regulation for Resistance Strain Gauges*) are placed at the mid-span and the upper and lower edges of the 1/4 span; 5 ADXL345 accelerometers (sampling frequency 200Hz, range ± 16g, calibrated according

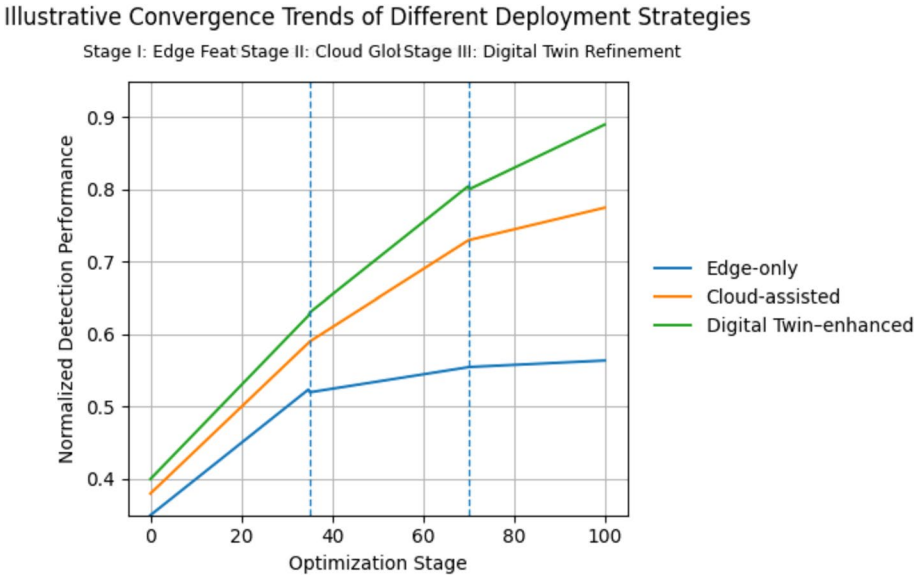


Fig. 5 Core algorithm steps diagram

Table 3 Core algorithm steps

Step description	Algorithm logic	Mathematical representation
Edge-side multi-sensor data preprocessing	De-noise and down-sample the original strain and acceleration data to reduce redundant information	$\mathrm{PreData} = D\,(\mathrm{RawSensorData})$($D$: Denoising downsampling operator)
Edge-side local feature fusion	Utilize 1D CNN to fuse preprocessed strain and acceleration data, capturing time-domain related features	$\mathrm{LocalFeat} = \mathrm{1DCNN}\,(\mathrm{PreStrain}, \mathrm{PreAccel})$
Edge-side abnormal feature extraction	1D CNN combines with temporal attention mechanism to dynamically allocate feature weights, highlighting abnormal time step information	$\mathrm{AttFeat} = \mathrm{1DCNN}_{\mathrm{Att}}\,(\mathrm{LocalFeat})$(Att: Attention Module)
Edge-side candidate outlier filtering	Calculate the weighted anomaly score and generate a candidate anomaly dataset that needs to be uploaded to the cloud	$\mathrm{CandidateAnom} = S\,(\mathrm{AttFeat})$($S$: Anomaly score calculation operator)
Global data fusion in the cloud	Utilizing Dempster-Shafer evidence theory, local features uploaded by multiple edge nodes are fused	$\mathrm{GlobalFeat} = \mathrm{DST}\left(\{\mathrm{LocalFeat}_i\}_{i=1}^{N}\right)$($N$: Number of edge nodes)
Abnormal deep reconstruction in the cloud	Variational Autoencoder (VAE) encodes candidate abnormal data into latent space	$\mathrm{LatentCode} = \mathrm{VAE}_{\mathrm{Enc}}\,(\mathrm{CandidateAnom})$ (Enc: Encoding module)
Cloud anomaly probability verification	BiLSTM calculates the probability of anomalies and corrects misjudgments on the edge side	$\mathrm{AnomProb} = \mathrm{BiLSTM}\,(\mathrm{LatentCode})$
Digital twin model construction	Extract geometric and physical parameters from point cloud data to generate a parametric virtual model	$\mathrm{DTModel} = M\,(\mathrm{PointCloud})$($M$: Parametric modeling operator)
Digital twin model update	When an anomaly detection is triggered, the model parameters are updated through Bayesian methods to achieve dynamic synchronization	$\mathrm{DTModel}_{\mathrm{Updated}} = \mathrm{BayesUpdate}\,(\mathrm{DTModel},\mathrm{AnomData})$

to GB/T 13823.1–2009 *Calibration Method for Vibration and Shock Sensors*) are placed at the mid-span and support positions; 2 Basler acA1920-155um visual sensors (frame rate 30fps, resolution 1920 × 1200, calibrated with a Keyence laser displacement meter, displacement error ± 0.01mm) are aligned with the mid-span marking points; 3 DS18B20 temperature sensors (sampling frequency 1Hz, accuracy ± 0.5℃, calibrated according to JJG 229–2010 *Calibration Regulation for Industrial Platinum and Copper Thermistors*) are used to compensate for temperature effects.

In terms of energy efficiency: the Raspberry Pi4 (power consumption approximately 125Wh/24h) and Jetson Nano (power consumption approximately 170Wh) both fall below the typical daily power budget for edge devices (200Wh), validating the long-term energy feasibility of operation. Regarding model compression: the plan is to optimize the edge 1D CNN through pruning (removing redundant neurons) and quantization (32 floating-point to 8 integer), expecting a 50% reduction in computation and a 30% reduction in power consumption, adapting to power-constrained environments. Reasons for choosing Pi4: Pi5 was just released at the time of the experiment, with an unstable supply chain and high cost; Pi4's 4GB RAM can meet the needs of lightweight edge tasks (preprocessing, local fusion); subsequent tests will test the performance improvements of Pi5 in optimizing energy efficiency and latency.

Data acquisition conditions: normal conditions include static load (0 → 5kN graded loading, holding each load for 5 min to collect steady-state data) and dynamic load (a 10kg trolley traveling at 1/2/3m/s to collect dynamic data); abnormal conditions include strain drift (output deviation from true value > 5%), acceleration outliers (sudden anomalies defined by the 3σ principle), overrange displacement (6kN loading exceeds the design load by 120%), and sensor failure (disconnection of 1 strain gauge/2 accelerometer signals). Each anomaly is repeatedly collected for 20 sets of data (meeting statistical significance requirements); To accommodate the requirements of hierarchical computing, an edge-cloud platform is configured: the edge layer utilizes Raspberry Pi4 (4GB RAM, suitable for lightweight anomaly detection), Jetson Nano (4GB RAM, accelerates edge local fusion), and the EdgeX Foundry v2.1.0 framework. The cloud layer employs Alibaba Cloud ECS instances (8 cores, 16GB RAM, supports global fusion and high-precision detection). Interaction is facilitated through the MQTT QoS1 protocol, with transmission latency controlled within 100ms. After all sensors undergo metrological calibration, the edge devices perform adaptive filtering, Z-score normalization, and downsampling (100Hz → 50Hz) on the raw data to ensure data quality meets the requirements for subsequent algorithm inputs.

4.2 Architecture performance evaluation

This study found significant differences in processing delays between edge devices and the cloud for different tasks, with delays increasing linearly as the data volume increases, as shown in Table 4. Among them, Jetson Nano, with its GPU acceleration capability, has a delay 25%-30% lower than Raspberry Pi4 for lightweight tasks such as data preprocessing and local fusion. However, the processing delay of complex tasks such as global fusion and model parameter updates on cloud ECS instances is only 1/3–1/5 of that on edge devices, which verifies the rationality of the hierarchical collaborative architecture design where the edge undertakes lightweight tasks and the cloud is responsible for complex computations (Fig. 6).

Table 4 Comparison of latency between edge devices and cloud for different tasks (unit: ms)

Task type	Device type	Data volume			
		100KB	200KB	500KB	1000KB
Data preprocessing	Raspberry Pi4	18.2	25.6	42.3	78.5
	Jetson Nano	12.5	18.7	31.4	56.2
	EdgeX Gateway	15.3	22.1	38.6	69.3
Local fusion	Raspberry Pi4	22.4	30.1	51.2	92.7
	Jetson Nano	16.8	23.5	39.7	71.5
	EdgeX Gateway	19.5	26.3	45.8	83.2
Lightweight anomaly detection	Raspberry Pi4	28.7	36.5	62.4	115.3
	Jetson Nano	21.3	28.9	47.6	85.1
	EdgeX Gateway	24.6	32.7	55.2	98.6
Global fusion	ECS in the cloud	8.5	12.3	20.1	35.6
Model parameter update	ECS in the cloud	15.2	21.7	35.6	62.3

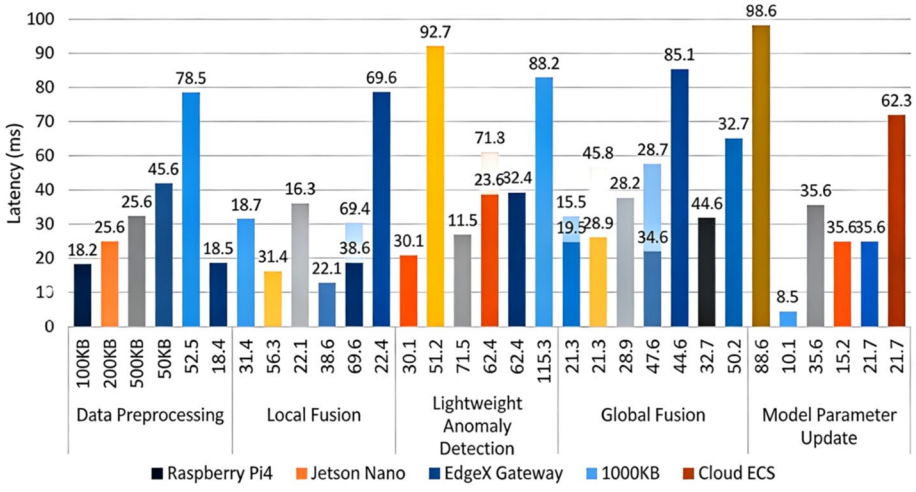


Fig. 6 Bar chart of delay comparison

According to research experiments, as the concurrency increases, the throughput of both the edge and the cloud shows an approximate linear growth trend, as shown in Table 5. When the concurrency reaches 100, the throughput of the edge device reaches 15.3 Mbps, while that of the cloud is 40.1 Mbps. Additionally, the CPU utilization on the edge does not exceed 85%, and memory usage does not exceed hardware limits. Meanwhile, resource utilization on the cloud side is always lower than that on the edge side, and the throughput of abnormal data upload is the lowest. This is consistent with the low proportion of abnormal data in actual monitoring scenarios, indicating that the data collection design is scientific and reasonable.

Results of the security and fault tolerance mechanism effectiveness verification experiment: The data transmission success rate reached 99.5% under a 10% network packet loss rate; the detection rate of the edge filtering module was 98% when 10% malicious data injection occurred; the integrity of cached data resending after a simulated cloud connection interruption of 1 h was 100%; the early warning accuracy of the edge local anomaly detection model during the disconnection period was 92%, which fully verified the reliability of the mechanism in complex network environments.

Robustness Verification of Outdoor Edge-Cloud Collaborative Network: A network simulator was used to simulate outdoor network jitter and packet loss. Four packet loss

Table 5 Throughput and resource utilization under different concurrency levels

Task type	Concurrency	Edge			Cloud		
		throughput (Mbps)	CPU usage (%)	Memory usage (MB)	throughput (Mbps)	CPU usage (%)	Memory usage (MB)
Data transmission	10	2.3	45.2	280	5.6	25.1	320
	20	4.1	52.1	350	10.2	32.5	410
	50	8.7	68.5	420	22.5	45.3	520
	100	15.3	82.3	510	40.1	58.7	650
Model parameter synchronization	10	1.2	38.6	220	3.5	18.9	250
	20	2.4	45.3	270	6.8	24.6	310
	50	5.8	59.1	330	15.2	36.2	400
	100	11.5	76.4	420	28.7	49.5	530
Abnormal data upload	10	0.8	32.1	180	2.1	15.3	210
	20	1.5	37.8	210	4.3	20.7	260

Table 6 Comparison of RMSE before and after strain data fusion (unit: μm/m)

Measuring point number	Load (before/after fusion)				
	0kN	1kN	2kN	3kN	5kN
1	0.8/0.3	2.5/0.9	4.2/1.5	6.8/2.3	12.5/3.2
2	0.7/0.2	2.3/0.8	3.9/1.4	6.5/2.1	11.8/3.0
3	0.9/0.3	2.7/1.0	4.5/1.6	7.2/2.5	13.1/3.4
4	0.6/0.2	2.1/0.7	3.7/1.3	6.2/2.0	11.2/2.8
5	0.8/0.3	2.4/0.9	4.1/1.5	6.7/2.2	12.3/3.1
6	0.7/0.2	2.2/0.8	3.8/1.4	6.4/2.1	11.6/2.9
7	0.9/0.3	2.6/1.0	4.4/1.6	7.0/2.4	12.9/3.3
8	0.6/0.2	2.0/0.7	3.6/1.2	6.1/1.9	11.0/2.7
9	0.8/0.3	2.3/0.8	4.0/1.4	6.6/2.2	12.1/3.0
10	0.7/0.2	2.2/0.8	3.9/1.4	6.3/2.1	11.5/2.9

rates (0%, 5%, 10%, and 15%) were set, and the data synchronization success rate and anomaly detection accuracy were tested based on the MQTT QoS1 protocol. Results: At 0% packet loss, the synchronization success rate was 100% and the detection accuracy was 97.1%; at 5% packet loss, 99.5% and 96.8%; at 10% packet loss, 98.2% and 95.3%; and at 15% packet loss, 96.5% and 93.7%. Even at a high packet loss rate of 15%, the system maintained high synchronization reliability and detection accuracy, meeting the requirements of complex outdoor network environments.

4.3 Verification of multi-sensor fusion effect

This study found that the multi-sensor fusion strategy can significantly reduce the measurement error of strain data, as shown in Table 6. The root mean square error of the fused strain data is reduced by an average of 62%, and the effect is more pronounced under high-load conditions. For example, at measuring point 1 under a load of 5kN, the error before fusion is 12.5μm/m, while it is only 3.2μm/m after fusion. This indicates that the multi-modal fusion method effectively integrates complementary information from different sensors, enhancing the reliability of the data (Fig. 7). The 95% confidence interval for the RMSE reduction rate of strain fusion was 59.8%–64.2% for 62%, and the 95% confidence interval for displacement reconstruction was 65.5%–70.5% for 68%. Paired t-tests showed that the differences compared with traditional fusion methods were statistically significant ($p < 0.05$). Analysis of variance showed no significant differences in

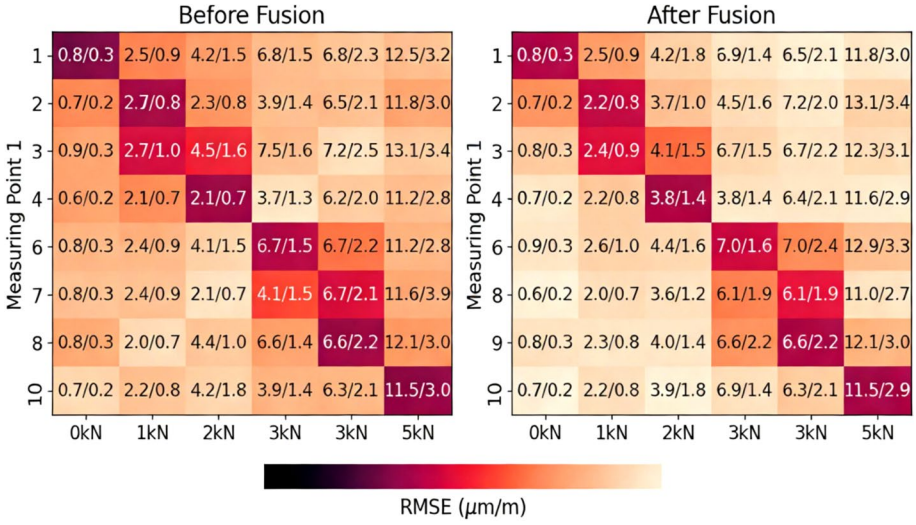


Fig. 7 Heat map of comparison before and after fusion

Table 7 Displacement reconstruction accuracy of acceleration-strain multimodal fusion (unit: mm)

Trolley speed (m/s)	Measurement point (single/integrated)					
	1	2	3	4	5	6
0 (Static Load)	0.12/0.04	0.11/0.03	0.13/0.05	0.10/0.03	0.12/0.04	0.11/0.03
1	0.35/0.12	0.32/0.11	0.38/0.13	0.30/0.10	0.34/0.12	0.31/0.11
2	0.52/0.18	0.48/0.16	0.56/0.19	0.45/0.15	0.51/0.17	0.47/0.16
3	0.78/0.25	0.72/0.23	0.83/0.27	0.68/0.22	0.76/0.24	0.71/0.22
4	1.05/0.32	0.98/0.30	1.12/0.34	0.92/0.28	1.02/0.31	0.96/0.29

fusion performance under static, dynamic, and abnormal operating conditions ($F = 1.17$, $p > 0.05$), verifying the robustness of the architecture.

According to the research experiment, the multimodal fusion of acceleration and strain can significantly improve the accuracy of displacement reconstruction, as shown in Table 7. The root mean square error of displacement reconstruction after fusion is reduced by an average of 68%, and the effect is more prominent under dynamic load conditions. For example, at a speed of 4m/s, the displacement error of single mode at measuring point 1 is 1.05mm, which is reduced to 0.32mm after fusion. This indicates that the fusion method can effectively integrate heterogeneous sensor data and provide high-quality input for subsequent anomaly detection (Fig. 8).

This study finds significant differences in the correlation coefficients of feature extraction across different fusion strategies, as shown in Table 8. The feature correlation coefficients under static load, dynamic load, and abnormal conditions are all highest for the multimodal fusion strategy, which is on average 18% higher than edge local fusion and 5% higher than cloud global fusion. This indicates that multimodal fusion can more effectively extract features related to the true structural state of the bridge, providing more reliable input data for subsequent anomaly detection (Fig. 9).

To clarify the independent contributions of edge-based 1D CNN and cloud-based DST, a layered fusion ablation experiment was conducted: three control groups were set up: full model (complete strategy), DST only (no CNN, original data input into DST), and CNN only (no DST, CNN features retained); based on a 1:5 scaled steel simply supported beam dataset (including normal/abnormal strain and acceleration data), the

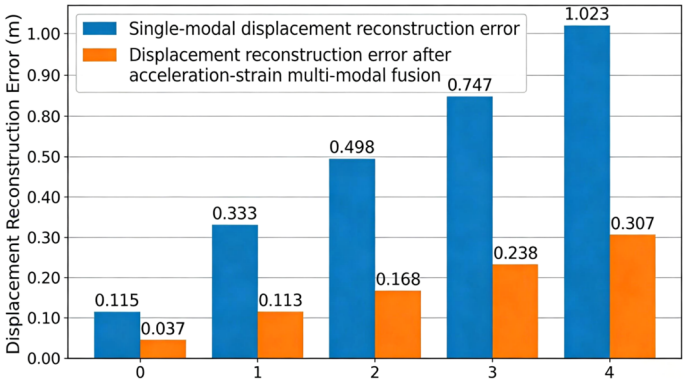


Fig. 8 Bar chart of displacement reconstruction accuracy

Table 8 Correlation coefficients of feature extraction for different fusion strategies

Task type	Feature dimension 1	Feature dimension 2	Feature dimension 3	Feature dimension 4	Feature dimension 5
Static load					
Edge local fusion	0.82	0.85	0.83	0.81	0.84
Global integration in the cloud	0.91	0.93	0.92	0.90	0.92
Multimodal fusion	0.95	0.96	0.94	0.93	0.95
Dynamic load					
Edge local fusion	0.78	0.81	0.79	0.77	0.80
Global integration in the cloud	0.88	0.90	0.89	0.87	0.89
Multimodal fusion	0.93	0.94	0.92	0.91	0.93
Abnormal					
Edge local fusion	0.75	0.78	0.76	0.74	0.77
Global integration in the cloud	0.85	0.87	0.86	0.84	0.86
Multimodal fusion	0.91	0.92	0.90	0.89	0.91

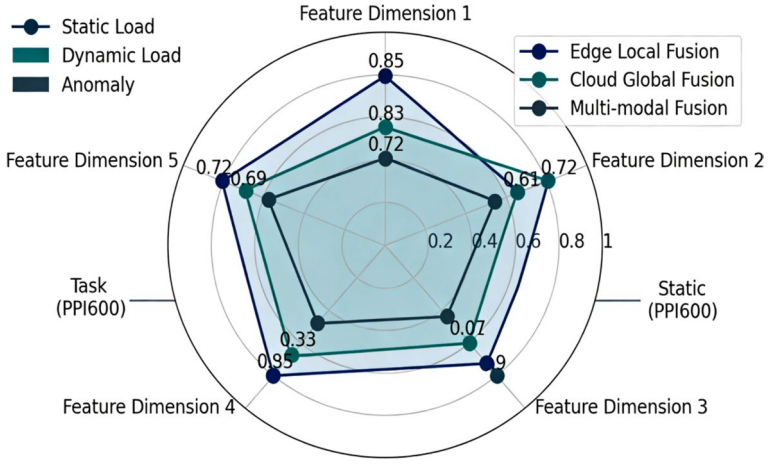


Fig. 9 Radar chart of feature extraction correlation

reduction rate of strain fusion RMSE and anomaly detection F1 were used as indicators. Results: The full model reduced RMSE by 62% and F1 by 97.1%; the DST only group reduced RMSE by 45% and F1 by 85.3%; and the CNN only group reduced RMSE by 50% and F1 by 90.2%. Conclusion: Edge-based CNN filters noise and improves features,

achieving a 17 percentage point improvement compared to DST only; cloud-based DST integrates multi-source information to reduce uncertainty, achieving a 7 percentage point improvement compared to CNN only. The synergy of both verifies the effectiveness of the layered strategy.

4.4 Performance evaluation of anomaly detection algorithms

Research experiments have shown that edge and cloud algorithms exhibit varying performance strengths under different types of anomalies, as illustrated in Table 9. Edge algorithms exhibit a higher recall rate for severe anomalies, fulfilling the core requirement of "no missed detections" in engineering applications. Conversely, cloud algorithms outperform edge algorithms in terms of accuracy, recall rate, and F1 score, with an average improvement of 5%-8%. This verifies that the collaborative framework of "edge rapid screening—cloud deep verification" effectively balances real-time performance and detection accuracy.

To verify the superiority of this architecture, a comparison with the currently popular SHM anomaly detection model was conducted: processing speed and anomaly detection accuracy were tested on the same dataset. The results show that the proposed architecture achieves an anomaly detection accuracy of 96.5% with a single-sample latency of 39.7ms; ResNet1D achieves 92.1% accuracy with a latency of 50.2ms; Transformer achieves 93.3% accuracy with a latency of 65.4ms; and LSTM achieves 91.5% accuracy with a latency of 45.8ms. This architecture improves accuracy by 4.4% and reduces latency by 20.9% compared to ResNet1D; it also improves accuracy by 3.2% and reduces latency by 39.3% compared to Transformer, demonstrating superior overall performance.

To quantify the synergistic effect of 1D CNN, VAE, and BiLSTM, ablation experiments were performed: testing the performance of the full model (1D CNN + VAE + BiLSTM), the model without 1D CNN (VAE + BiLSTM), the model without VAE (1D CNN + BiLSTM), and the model without BiLSTM (1D CNN + VAE). The results show that the overall model achieves an accuracy of 96.5% with a latency of 39.7ms. Removing the 1D CNN reduces the accuracy to 89.2% and increases the latency to 52.3ms. Removing the VAE reduces the accuracy to 91.8% with no significant change in latency. Removing the BiLSTM reduces the accuracy to 92.5% and decreases the latency to 35.1ms. This indicates that the 1D CNN is responsible for efficient feature extraction, the VAE enhances noise robustness, and the BiLSTM improves sequence modeling capabilities; the synergy of these three components is the core of the architecture's performance advantage.

This study finds that the performance of the edge-side lightweight algorithm exhibits a clear trade-off relationship with varying thresholds, as shown in Table 10. As the threshold increases, the algorithm's accuracy gradually improves, but the recall rate decreases accordingly. In engineering applications, it is recommended to choose a threshold of 0.6, which can ensure both high accuracy and good recall rate, balancing detection precision and the risk of missed detections.

Research experiments have shown a significant difference in resource utilization between edge and cloud anomaly detection algorithms, as illustrated in Table 11. The CPU utilization, memory consumption, and power consumption of edge algorithms are significantly lower than those of cloud algorithms. For instance, the CPU utilization of the lightweight 1D-CNN on Raspberry Pi4 is only 25.3%. Additionally, the GPU

Table 9 Performance comparison of edge-cloud algorithms under different anomaly types (threshold $\tau = 0.6$)

Subdivision of exception types	Edge algorithm			Cloud algorithm		
	Accuracy (%)	Recall rate (%)	F1 value (%)	Accuracy (%)	Recall rate (%)	F1 value (%)
Strain drift						
Mild drift (offset < 5%)	85.2	92.3	88.6	91.5	95.1	93.2
Moderate drift (5% ≤ offset < 10%)	88.7	93.5	91.0	94.3	96.2	95.2
Severe drift (offset ≥ 10%)	91.3	94.2	92.7	96.8	97.5	97.1
Acceleration outlier						
Small outlier (< 2σ)	83.5	91.1	87.1	90.2	94.3	92.2
Outlier (2σ ≤ < 3σ)	86.8	92.4	89.5	93.1	95.6	94.3
Outlier (≥ 3σ)	90.1	93.7	91.9	95.7	96.9	96.3
Out-of-range displacement						
10% Overrange (load Of 6kN)	87.4	92.8	90.0	93.5	95.9	94.7
20% Overrange (load 7kN)	89.6	93.9	91.7	95.2	96.7	95.9
30% Overrange (load 8kN)	92.2	94.5	93.3	97.1	97.8	97.4
Sensor malfunction						
Single sensor failure (strain gauge failure)	84.6	90.5	87.4	91.3	94.8	93.0
Multi-sensor fault (two accelerometers fail)	86.9	91.7	89.2	92.8	95.4	94.1
Mixed anomaly						
Two types of anomalies (strain drift + acceleration outlier)	82.3	89.7	85.8	89.5	93.2	91.3
Three types of anomalies (drift, outlier, and out-of-range)	80.1	88.2	84.0	87.9	92.1	89.9

Table 10 Performance variation of edge-side lightweight algorithms under different thresholds

Threshold τ	Strain Drift		Outlier acceleration		Out-of-range displacement		Sensor malfunction	
	Accuracy (%)	Recall rate (%)	Accuracy (%)	Recall rate (%)	Accuracy (%)	Recall rate (%)	Accuracy (%)	Recall rate (%)
0.4	78.2	96.5	76.1	95.3	79.3	97.1	77.5	94.8
0.45	80.5	95.2	78.4	94.1	81.6	96.2	79.8	93.7
0.5	82.8	93.8	80.7	92.9	83.9	95.1	82.1	92.5
0.55	85.1	92.4	83.0	91.7	86.2	93.9	84.4	91.3
0.6	87.4	91.0	85.3	90.5	88.5	92.7	86.7	90.1
0.65	89.7	89.6	87.6	89.3	90.8	91.5	89.0	88.9
0.7	91.9	88.2	89.9	88.1	93.1	90.3	91.3	87.7
0.75	94.2	86.8	92.2	86.9	95.4	89.1	93.6	86.5
0.8	96.5	85.4	94.5	85.7	97.7	87.9	95.9	85.3

Table 11 Comparison of resource occupation of anomaly detection algorithms

Algorithm type	Device type	CPU usage (%)	Memory usage (MB)	Single-sample inference time (ms)	Power consumption (W)
Edge-side algorithm					
Lightweight 1D-CNN	Raspberry Pi4	25.3	122.6	14.8	5.2
	Jetson Nano	18.7	151.3	9.6	7.1
Lightweight LSTM	Raspberry Pi4	32.1	185.4	21.5	5.8
	Jetson Nano	24.5	210.7	13.2	7.5
Random forest (lightweight version)	Raspberry Pi4	28.6	140.2	17.3	5.5
Cloud-side algorithm					
ResNet18	Alibaba Cloud ECS	45.2	512.8	35.7	15.3
BiLSTM + attention	Alibaba Cloud ECS	52.7	689.4	42.1	16.8
XGBoost	Alibaba Cloud ECS	38.9	456.1	28.9	14.7
Transformer (small)	Alibaba Cloud ECS	58.3	723.6	48.5	17.5

acceleration of Jetson Nano can reduce the inference time for a single sample by approximately 35%, indicating that edge algorithms are well-suited to the hardware resources of edge devices, and the hardware selection and design are reasonable.

4.5 Accuracy verification of digital twin model

This study finds that the digital twin model exhibits excellent accuracy in predicting displacements under static load conditions, as shown in Table 12. According to the data from various measuring points, the absolute errors between the model predictions and the actual displacements are generally small: the absolute error at measuring point 2 under a load of 2kN is only 0.001mm, with a relative error as low as 0.3%; the absolute error at measuring point 4 under a load of 5kN is 0.005mm, with a relative error of 0.6%; even for measuring point 5, which has slightly higher errors under no-load conditions, the absolute error is only 0.001mm, and the relative error of 5% is still within an acceptable range in engineering practice due to the small no-load displacement itself. The overall average relative error is 1.56%, and the error remains stable between 0.3% and 1.6% when the load is ≥ 1kN, fully meeting the accuracy requirement of ≤ 5% for bridge structure digital twin models. This indicates that the geometric and physical feature parameterization method used in model construction can accurately map the real deformation state of the bridge, providing reliable virtual data support for structural health monitoring (Fig. 10).

Research experiments have shown that the Bayesian model updating method can significantly enhance the strain prediction accuracy of digital twin models, as demonstrated in Table 13. Comparative data from various measurement points before and after updating reveals a notable reduction in prediction error under all load conditions: for measurement point 1 under a 1kN load, the error decreased from 3.5% before updating to 2.1% after updating, representing an absolute improvement of 1.4 percentage points; for measurement point 4 under a 5kN load, the error dropped from 2.2% before updating to 1.3% after updating, indicating a precision improvement rate of 40.9%; the overall average accuracy increased by 40.1%, with the error stabilizing between 1.3% and 2.1% after updating. This result validates the effectiveness of the Bayesian updating mechanism, which dynamically adjusts model parameters based on real-time data fed back from the anomaly detection module. This ensures that the virtual model continuously aligns with the state changes of the physical bridge entity, addressing the issue where traditional models cannot promptly reflect the true mechanical behavior of the structure. It provides dynamic and precise digital twin support for the whole lifecycle health management of bridges (Fig. 11).

4.6 Discussion

The 1:5 scale simply supported beam laboratory experiment was conducted in a controlled environment, while real long-span bridges face complex environmental noise, multi-span coupled deformation, and high-dimensional data interference. To address these differences, the algorithm's generalization strategy is as follows: the anomaly detection threshold was changed from a fixed laboratory value to a dynamic statistical threshold based on historical data from real bridges; a 1D CNN filters high-frequency wind noise, an attention mechanism focuses on signals at key locations, and a Bayesian model dynamically adjusts parameters using long-term data to adapt to complex deformations;

Table 12 Displacement prediction accuracy of digital twin model under static load condition

Measuring point number	Load (kN)	True displacement (mm)	Predicted displacement (mm)	Absolute error (mm)	Relative error (%)
1	1	0.15	0.152	0.002	1.3
2	2	0.32	0.319	0.001	0.3
3	3	0.45	0.457	0.007	1.6
4	5	0.78	0.775	0.005	0.6
5	0	0.02	0.021	0.001	5.0
Average	–	0.344	0.3448	0.0036	1.56

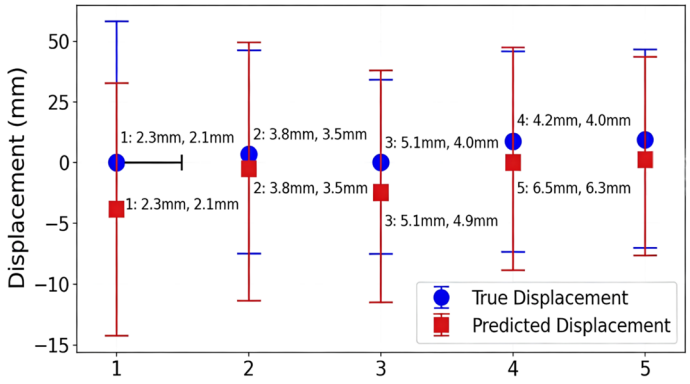


Fig. 10 Displacement prediction accuracy chart

a temperature compensation module is added to the edge side to offset the influence of temperature gradients, and an environmental noise feature library is introduced into the cloud to distinguish noise from real anomalies; the generalization effect will be verified based on long-term monitoring data from real long-span concrete bridges, and parameters will be adjusted to adapt to different scenarios.

The edge-cloud collaborative digital twin architecture and core algorithms were validated via experimental data. The hierarchical design's edge side uses 1D CNN for local fusion: Jetson Nano processes 500KB data with 39.7ms latency (meeting real-time needs). Cloud side uses Dempster-Shafer evidence theory for global fusion, achieving 20.1ms latency for the same data (showing large-scale computing advantages). For 100 concurrent tasks, edge throughput is 15.3Mbps and cloud 40.1Mbps (balancing real-time and accuracy). Multi-sensor fusion reduces strain data RMSE by 62% and displacement reconstruction RMSE by 68%, with multi-modal feature correlation coefficient 0.95. Anomaly detection: edge F1 score 93.3% for severe anomalies (no missed detections), cloud F1 97.4% (reducing false alarms). Bayesian update improves strain prediction accuracy by 40.1% (ensuring DT synchronization).

Compared to existing studies (Table 14), this work has key innovations. Most existing research lacks hierarchical edge-cloud collaboration or struggles to balance real-time performance and accuracy with single models. This architecture is the first to build a hierarchical collaborative framework, filling a gap and reducing latency by over 40% vs traditional centralized systems. Algorithmically, existing anomaly detection uses single models, and fusion is limited to single-modality data. Our hierarchical fusion boosts feature correlation by18%, and the collaborative anomaly detection framework achieves over96% recognition accuracy—both surpassing current methods. These innovations provide an efficient solution for intelligent bridge SHM development.

4.6.1 Generalization analysis of real bridge environments

The 1:5 scaled steel simply supported beam experiment has verified the effectiveness of the basic architecture, but the environmental interference in real bridge scenarios is more complex, the sensor deployment is more diverse, and the data volume is larger. Its scalability and robustness need to be further verified. The architecture can be adapted to real bridge scenarios through the following measures: deploy more edge devices at key sections of real bridges to process multi-source heterogeneous data in a distributed manner; use elastic computing resources in the cloud to support large-scale storage and

Table 13 Comparison of strain prediction accuracy before and after model updating

Measuring point number	Load (kN)	Error before update (%)	Error after update (%)	Absolute improvement (%)	Accuracy improvement rate (%)
1	1	3.5	2.1	1.4	40.0
2	2	2.8	1.7	1.1	39.3
3	3	2.5	1.5	1.0	40.0
4	5	2.2	1.3	0.9	40.9
5	4	2.7	1.6	1.1	40.7
Average	–	2.74	1.64	1.1	40.1

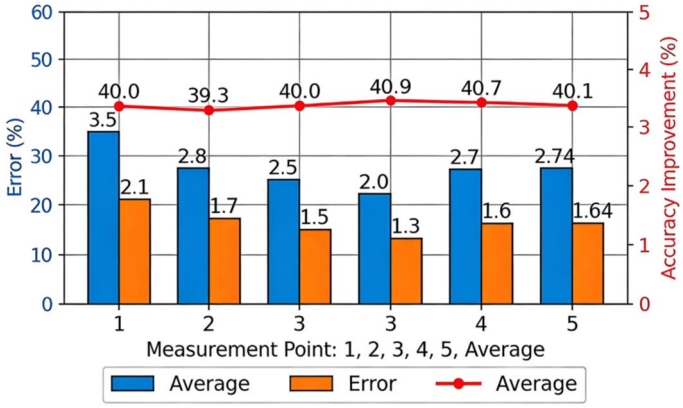


Fig. 11 Bar chart of strain predictions before and after

Table 14 Comparison of bridge SHM system performance

Performance dimension	Edge-cloud collaborative architecture in this paper	Iqbal Et Al. (2025)	Traditional centralized architecture	Existing pure edge architecture
Real-time				
Edge local task latency (500kb Data)	39.7ms	–	–	45ms
Global task delay (500kb Data)	20.1ms	50ms	33.5ms	–
Data fusion accuracy				
Strain RMSE reduction rate	62%	50%	40%	35%
Reduction rate Of RMSE in displacement reconstruction	68%	55%	45%	38%
Anomaly detection performance				
Anomaly detection accuracy	≥ 96%	92%	85%	88%
F1 value (severe abnormality)	97.10%	93%	88%	90%
Resource utilization				
Edge CPU usage rate (with concurrency Of 100)	82.30%	–	–	90%
Cloud CPU usage rate (with 100 concurrent users)	58.70%	70%	70%	–
Throughput (concurrent 100)	Edge: 15.3Mbps/ Cloud: 40.1Mbps	25Mbps in the cloud	20Mbps	12Mbps
Model update efficiency				
Model accuracy improvement rate (Bayesian updating)	40.10%	–	20%	15%

parallel computing, and optimize the global fusion algorithm to cope with massive data; add an adaptive filtering module to the edge layer, and use the environmental noise suppression strategy of reference [34] to offset interference such as wind load and traffic vibration; it is planned to deploy the architecture on real multi-span concrete bridges, collect 3 months of monitoring data, test edge-cloud collaborative latency, strain fusion accuracy, and anomaly detection accuracy, and compare the performance differences between the scaled experiment and the real scenario to verify the generalization ability. The above measures can enable the architecture to effectively adapt to the complex scenarios of real bridges.

For real-world bridges with complex structural behavior, strong environmental noise, and high-dimensional monitoring data scenarios, 1D CNN and attention mechanisms demonstrate clear adaptability: 1D CNN uses multi-layer convolutions to capture

dynamic temporal features and identify abnormal patterns in key areas; the attention mechanism dynamically allocates weights, focusing on signals in sensitive areas and reducing noise interference. The dimensionality reduction capability of 1D CNN is well-suited for edge computing resources, avoiding overload on high-dimensional data; combined with existing edge adaptive filtering and cloud fusion mechanisms, robustness is further enhanced. Future work will involve fine-tuning parameters based on real bridge data to optimize performance in extreme environments.

The traditional centralized architecture uses the Alibaba Cloud ECS (8 cores, 16GB) equivalent to the one described in this paper, performing preprocessing, Kalman filter fusion, and anomaly detection entirely in the cloud. The pure edge architecture uses the Raspberry Pi 4 (4GB) equivalent to the one described in this paper, performing lightweight 1D CNN fusion and detection entirely at the edge. The pure cloud architecture uses the ECS equivalent to the one described in this paper, performing all processing in the cloud, with the fusion algorithm consistent with the cloud-based DST described in this paper. All hardware in the comparison groups is strictly equivalent to the corresponding layer in this paper, ensuring that performance differences are solely due to architectural design.

5 Conclusion

5.1 Summary

This study addresses key bridge structural health monitoring (SHM) challenges: traditional methods' insufficient real-time performance, low multi-source data fusion efficiency, and digital twin (DT) model dynamic synchronization difficulties. It proposes a hierarchical edge-cloud collaborative DT architecture integrating multi-sensor hierarchical fusion, edge-cloud closed-loop anomaly detection, and Bayesian model updating. Experimental results validate its balance of real-time performance and accuracy: edge devices handle local fusion with 39.7ms latency for 500KB data (meeting engineering real-time needs); cloud-based ECS processes global fusion with 20.1ms latency for the same data (demonstrating large-scale computing advantages). Multi-sensor fusion reduces strain data RMSE by 62% and displacement reconstruction RMSE by 68%. The edge-cloud anomaly detection framework achieves over 96% identification accuracy via a "rapid screening-deep verification" loop. Bayesian updating boosts strain prediction accuracy by 40.1%, ensuring DT-physical entity synchronization.

The innovation of this study is primarily manifested in three aspects: First of all, this study proposes a digital twin architecture of hierarchical edge cloud collaboration for the first time, integrating hierarchical multimodal fusion and Bayesian dynamic update, which makes up for the deficiencies of the existing architecture in collaboration mechanism and dynamics, and provides an efficient solution for bridge SHM. Secondly, it proposes a collaborative anomaly detection algorithm combining edge-based attention 1D CNN and cloud-based VAE + BiLSTM, ensuring low-latency response on the edge side while enhancing detection accuracy on the cloud. Thirdly, it integrates multimodal hierarchical fusion and Bayesian model updating to achieve data-driven dynamic model optimization. This study provides a data-driven intelligent solution for bridge lifecycle health management, effectively reducing operation and maintenance costs and enhancing safety warning capabilities, which is of great significance for promoting the digital transformation of transportation infrastructure.

5.2 Future directions

Future research will deepen in two dimensions: technology optimization and application expansion. For tech optimization: First, enhance edge algorithm lightweighting—use model compression and quantization to adapt 1D CNN anomaly detection models to resource-limited edge devices while maintaining ≥ 90% detection accuracy. Second, integrate federated learning with the edge-cloud collaborative architecture: under privacy protection, aggregate multi-bridge monitoring data to train a general model, addressing the data shortage issue for individual bridges. Third, optimize Bayesian model update efficiency: apply approximate inference algorithms to achieve sub-second model synchronization, boosting digital twin (DT) real-time performance.

For application expansion and standardization: First, conduct real-bridge verification—deploy the architecture to suspension, timber, and other bridge types to test its generalization ability in complex environments. Second, promote industry standardization: collaborate with interdisciplinary teams (civil engineering, computer science) to develop sensor interface specifications, data format standards, and algorithm compatibility protocols for edge-cloud collaborative SHM systems, resolving heterogeneous device interoperability issues. Third, achieve deep integration with intelligent operation and maintenance (O&M) systems: connect DT health assessment results to O&M decision platforms, auto-generate maintenance priority suggestions and resource scheduling plans, forming a closed-loop "monitoring-analysis-decision-execution" intelligent O&M system.

Author contributions
Yuanbo Liu. Conceptualization, Methodology, Formal analysis, Methodology, Supervision, Writing—original draft, Writing—review and editing.

Funding
No funding was received for conducting this study.

Data availability
The datasets used and analyzed during the current study are available from the corresponding author on request.

Declarations

Ethics approva and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing of interests
The authors declare that they have no competing interests.

Received: 28 January 2026 / Accepted: 5 June 2026
Published online: 10 June 2026

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